

CYBERDEFENDERS LAB REPORT

PsExec Hunt

SMB Lateral Movement and Remote Service Execution

Network Forensics Investigation

Report Field	Value
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Platform	CyberDefenders
Track	SOC Analyst Tier 1
Evidence	psexec-hunt.pcapng
Primary Tool	Wireshark
Report Date	1 October 2026
Timezone	UTC

Handling statement: This report analyzes a training PCAP. Conclusions are restricted to evidence visible in the supplied capture and are not presented as claims about a production environment.

SMB2 • NTLMSSP • DCE/RPC • SVCCTL • PsExec

Executive Summary

The supplied packet capture records a PsExec-style lateral-movement sequence originating from HR-PC (10.0.0.130). The host authenticated to SALES-PC (10.0.0.133) over SMB using the account ssales, accessed IPC\$ and ADMIN\$, transferred PSEXESVC.exe, created and started a remote Windows service through SVCCTL, and established named pipes for standard input, output, and error.

Approximately four minutes later, the same source initiated SMB connections to 10.0.0.131. The target's NTLM challenge identified that system as MARKETING-PC, establishing evidence of further attempted or completed lateral movement.

The capture proves the authentication, file transfer, service-control sequence, and named-pipe exchange. It does not reveal how the credentials were obtained, and the observed PsExec stdout payload cannot be decoded as readable command output from the PCAP alone.

Key Findings

Category	Finding
Origin	10.0.0.130 / HR-PC
First pivot	10.0.0.133 / SALES-PC
Credential	ssales via NTLM
Execution	PSEXESVC.exe through a temporary Windows service
Shares	ADMIN\$ for deployment; IPC\$ for RPC and named pipes
Further pivot	10.0.0.131 / MARKETING-PC

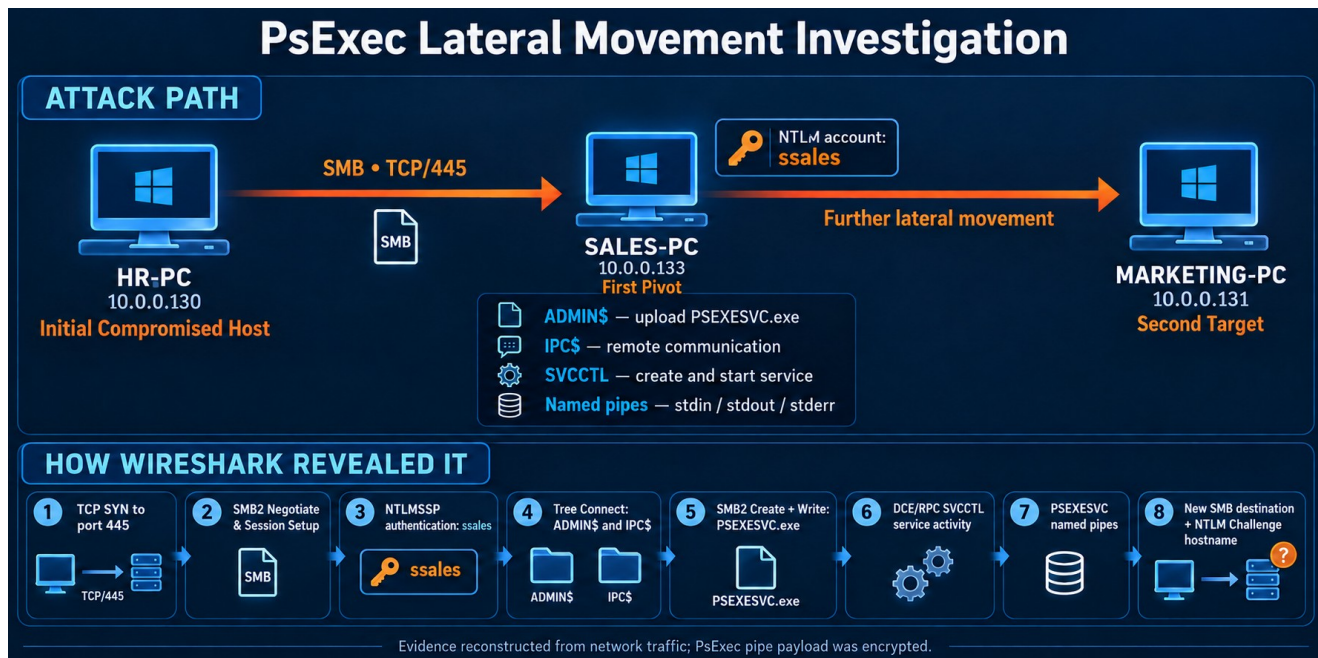


Figure 1. Visual summary of the reconstructed PsExec lateral-movement chain.

1. Scope and Objectives

The objective was to reconstruct the sequence visible in the PCAP and answer the investigation questions without relying on packet volume alone or assuming that every use of PsExec is malicious.

- Identify the machine from which the suspicious SMB activity originated.
- Determine the hostname of the first pivot target.
- Recover the username used during NTLM authentication.
- Identify the remote service executable.
- Determine the shares used for installation and communication.
- Identify the second host targeted for lateral movement.
- Map the observed behaviors to MITRE ATT&CK and document defensive actions.

2. Evidence and Tooling

Item	Details
Evidence	psexec-hunt.pcapng
Packet count	40,294
Tool	Wireshark
Primary protocols	SMB2, NTLMSSP, DCE/RPC, SVCCTL
Service	SMB over TCP/445
Analysis timezone	UTC

3. Investigation Approach

1. Establish a protocol baseline and prioritize SMB-oriented traffic.
2. Keep client SMB negotiation requests to identify initiators and targets.
3. Use NTLM Session Setup messages to recover server, client, and account identities.
4. Trace Tree Connect, Create, Write, Read, and DCE/RPC operations in the relevant conversation.
5. Correlate file transfer, service creation, service start, and named-pipe activity.
6. Search for later TCP/445 connections from the same source and identify additional targets.

Analytical principle: a single artifact can be benign. The conclusion depends on the correlated sequence: valid-account authentication, administrative-share access, service binary transfer, SVCCTL execution, and PsExec named pipes.

4. Detailed Packet Analysis

4.1 Protocol Baseline

Wireshark's Protocol Hierarchy Statistics showed that SMB2 accounts for approximately 93.5% of the 40,294 packets. DCE/RPC and SVCCTL are also present. This combination is consistent with Windows file sharing plus remote service management and therefore provides a logical starting point.

Statistics → Protocol Hierarchy

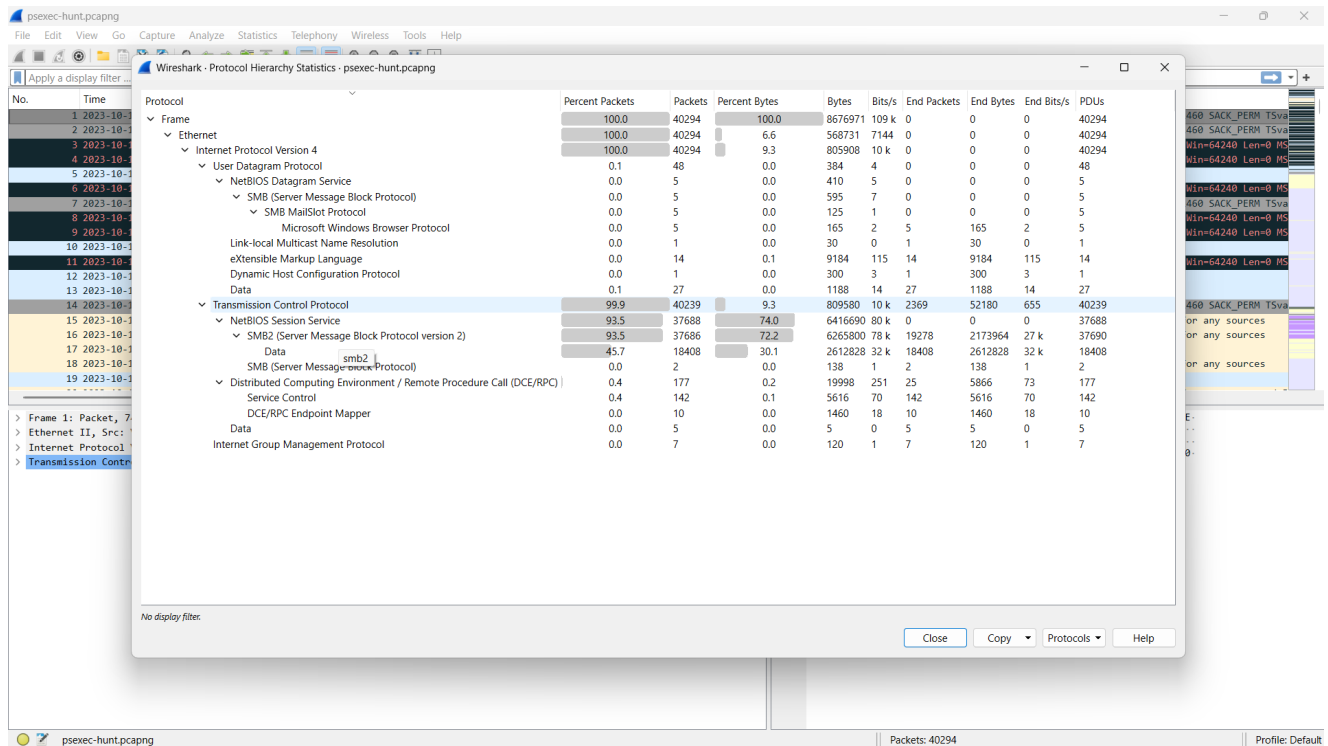


Figure 2. Protocol hierarchy showing SMB2 dominance and DCE/RPC service-control traffic.

4.2 Initial SMB Source and Targets

The following filter keeps SMB2 NEGOTIATE requests and removes responses. For a request, the Source column identifies the initiating client.

```
smb2.cmd == 0 && smb2.flags.response == 0
```

The earliest relevant request originates from 10.0.0.130 and targets 10.0.0.133 on TCP/445. Later requests from the same source target 10.0.0.131.

The screenshot shows the PSEXEC HUNT application window. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons. The main window displays a list of network packets filtered by the expression `smb2.cmd == 0 && smb2.flags.response == 0`. The list shows four SMB2 negotiate requests from 10.0.0.130 to 10.0.0.133 and 10.0.0.131.

No.	Time	Source	Destination	Protocol	Length	Info
128	2023-10-11 07:42:08.861351696Z	10.0.0.130	10.0.0.133	SMB2	286	Negotiate Protocol Request
38511	2023-10-11 07:46:19.909032675Z	10.0.0.130	10.0.0.131	SMB2	286	Negotiate Protocol Request
38511	2023-10-11 07:46:21.972971193Z	10.0.0.130	10.0.0.131	SMB2	302	Negotiate Protocol Request
39108	2023-10-11 07:46:33.559488606Z	10.0.0.130	10.0.0.131	SMB2	302	Negotiate Protocol Request
39696	2023-10-11 07:46:40.831640917Z	10.0.0.130	10.0.0.131	SMB2	302	Negotiate Protocol Request

The bottom pane shows a detailed analysis of the selected packet (Frame 128). It includes the following information:

- Frame 128: Packet, 286 bytes on wire (2288 bits), 286 bytes captured (2288 bits) on interface ens33, id 0
- Ethernet II, Src: VMware_06:cc:75 (00:0c:29:06:cc:75), Dst: VMware_59:23:50 (00:0c:29:59:23:50)
- Internet Protocol Version 4, Src: 10.0.0.130, Dst: 10.0.0.133
- TCP, Src Port: 49696, Dst Port: 445, Seq: 74, Ack: 453, Len: 232
- NetBIOS Session Service
- SMB2 (Server Message Block Protocol version 2), Negotiate Protocol Request, MessageId 1

The packet details pane also shows the raw packet data in hexadecimal and ASCII format.

Figure 3. Client-originated SMB2 negotiation requests from 10.0.0.130.

4.5 Administrative and IPC Share Access

SMB2 command 3 is TREE_CONNECT. The client used it to request both ADMIN\$ and IPC\$ on SALES-PC.

```
ip.stream eq 7 && smb2.cmd == 3
```

4.5.1 ADMIN\$ — Service Deployment

ADMIN\$ maps to the remote Windows directory and provides the location used by PsExec to deploy its temporary service executable.

The image displays a Wireshark network capture of SMB traffic. The top pane shows a list of packets, with packet 138 selected. The middle pane shows the packet details for the selected packet, and the bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
134	2023-10-11 07:42:08.881275840Z	10.0.0.130	10.0.0.133	SMB2	164	Tree Connect Request, Tree: '\\10.0.0.133\IPC\$'
135	2023-10-11 07:42:08.881481405Z	10.0.0.133	10.0.0.130	SMB2	138	Tree Connect Response, Tree: '\\10.0.0.133\IPC\$'
138	2023-10-11 07:42:08.882995105Z	10.0.0.130	10.0.0.133	SMB2	168	Tree Connect Request, Tree: '\\10.0.0.133\ADMIN\$'
139	2023-10-11 07:42:08.883428516Z	10.0.0.133	10.0.0.130	SMB2	138	Tree Connect Response, Tree: '\\10.0.0.133\ADMIN\$'
38470	2023-10-11 07:45:56.318784065Z	10.0.0.130	10.0.0.133	SMB2	168	Tree Connect Request, Tree: '\\10.0.0.133\ADMIN\$'
38471	2023-10-11 07:45:56.319726460Z	10.0.0.133	10.0.0.130	SMB2	138	Tree Connect Response, Tree: '\\10.0.0.133\ADMIN\$'

The packet details for packet 138 (SMB2 Tree Connect Request) are as follows:

- Frame 138: Packet, 168 bytes on wire (1344 bits), 168 bytes captured (1344 bits) on interface ens33, id 0
- Ethernet II, Src: VMware_06:cc:75 (00:0c:29:06:cc:75), Dst: VMware_59:23:50 (00:0c:29:59:23:50)
- Internet Protocol Version 4, Src: 10.0.0.130, Dst: 10.0.0.133
- Transmission Control Protocol, Src Port: 49696, Dst Port: 445, Seq: 1247, Ack: 1873, Len: 114
- NetBIOS Session Service
- SMB2 (Server Message Block Protocol version 2), Tree Connect Request, MessageId 6
 - SMB2 Header
 - Tree Connect Request (0x03)
 - StructureSize: 0x0009
 - Flags: 0x0000
 - Tree: '\\10.0.0.133\ADMIN\$'

The raw packet data (hex dump) is as follows:

```

0000  00 0c 29 59 23 50 00 0c 29 06 cc 75 00 00 45 00  ...YMP...u..E-
0010  00 9a 03 cd 40 00 80 06 e1 8a 0a 00 00 82 0a 00  ...@...
0020  00 85 c2 20 01 bd dd ef 93 3e d8 89 67 36 50 18  ...>gD-
0030  20 14 66 d2 00 00 00 00 00 6e fe 53 4d 42 40 00  ...f.....nSMB-
0040  01 00 00 00 00 03 00 01 00 18 00 00 00 00 00 00  ...
0050  00 00 06 00 00 00 00 00 00 00 ff fe 00 00 00 00  ...
0060  00 00 39 00 00 00 00 30 00 00 47 f1 c8 bd ba 35  ...9...0...6...6
0070  45 9d 4a 6b 59 4a 25 18 e3 17 09 00 00 00 48 00  ...E-3KY3K...H-
0080  26 00 5c 00 5c 00 31 00 30 00 2e 00 30 00 2e 00  ...&-1-1-0...
0090  30 00 2e 00 31 00 33 00 33 00 5c 00 41 00 44 00  ...0...1-3-3-A-D-
00a0  4d 00 49 00 4e 00 24 00  ...M-I-N-$-
  
```

Figure 6. SMB Tree Connect request to the ADMIN\$ administrative share.

4.5.2 IPC\$ — Control and Communication

IPC\$ supports inter-process communication, RPC endpoints, and named-pipe access used by PsExec for control and command I/O.

The image shows a Wireshark capture of network traffic from a file named 'psexec-hunt.pcapng'. The filter is 'ipstream eq 7 & smb2.cmd == 3'. The packet list shows several SMB2 Tree Connect requests and responses between 10.0.0.130 and 10.0.0.133. Packet 134 is selected, showing details for an SMB2 Tree Connect Request (MessageId 4) to the '\\10.0.0.133\\IPC\$' tree. The packet bytes pane shows the raw data of the request.

No.	Time	Source	Destination	Protocol	Length	Info
134	2023-10-11 07:42:08.881275840Z	10.0.0.130	10.0.0.133	SMB2	164	Tree Connect Request, Tree: '\\10.0.0.133\\IPC\$'
135	2023-10-11 07:42:08.881481405Z	10.0.0.133	10.0.0.130	SMB2	138	Tree Connect Response, Tree: '\\10.0.0.133\\IPC\$'
138	2023-10-11 07:42:08.882995105Z	10.0.0.130	10.0.0.133	SMB2	168	Tree Connect Request, Tree: '\\10.0.0.133\\ADMIN\$'
139	2023-10-11 07:42:08.883428516Z	10.0.0.133	10.0.0.130	SMB2	138	Tree Connect Response, Tree: '\\10.0.0.133\\ADMIN\$'
38470	2023-10-11 07:45:56.318784065Z	10.0.0.130	10.0.0.133	SMB2	168	Tree Connect Request, Tree: '\\10.0.0.133\\ADMIN\$'
38471	2023-10-11 07:45:56.319726460Z	10.0.0.133	10.0.0.130	SMB2	138	Tree Connect Response, Tree: '\\10.0.0.133\\ADMIN\$'

Frame 134: Packet, 164 bytes on wire (1312 bits), 164 bytes captured (1312 bits) on interface ens33, id 0
 > Ethernet II, Src: VMware_06:cc:75 (00:0c:29:06:cc:75), Dst: VMware_59:23:50 (00:0c:29:59:23:50)
 > Internet Protocol Version 4, Src: 10.0.0.130, Dst: 10.0.0.133
 > Transmission Control Protocol, Src Port: 49696, Dst Port: 445, Seq: 1013, Ack: 1369, Len: 110
 > NetBIOS Session Service
 > SMB2 (Server Message Block Protocol version 2), Tree Connect Request, MessageId 4
 > SMB2 Header
 > Tree Connect Request (0x03)
 > StructureSize: 0x0009
 > Flags: 0x0000
 > Tree: '\\10.0.0.133\\IPC\$'

Figure 7. SMB Tree Connect activity showing IPC\$ communication access.

4.6 PSEXESVC.exe Creation and Transfer

SMB2 command 5 is CREATE. In SMB, CREATE can open or create a file, directory, or named pipe. The request identifies PSEXESVC.exe.

```
ip.stream eq 7 && smb2.cmd == 5
```

The screenshot shows a Wireshark capture of network traffic. The packet list on the left shows a sequence of SMB2 messages. Packet 144 is selected, showing a Create Request (SMB2 cmd 5) for the file 'PSEXESVC.exe'. The packet details pane on the right shows the structure of the Create Request, including the file name, file attributes, and share access. The packet bytes pane at the bottom shows the raw hex and ASCII data of the packet.

Figure 8. SMB2 Create request for PSEXESVC.exe.

4.6.1 Transfer Confirmation with SMB Write

A file name alone is insufficient to prove transfer. SMB2 command 9 is WRITE; repeated client writes, including large chunks, confirm that executable content was copied to the target.

```
ip.stream eq 7 && smb2.cmd == 9 && smb2.flags.response == 0
```

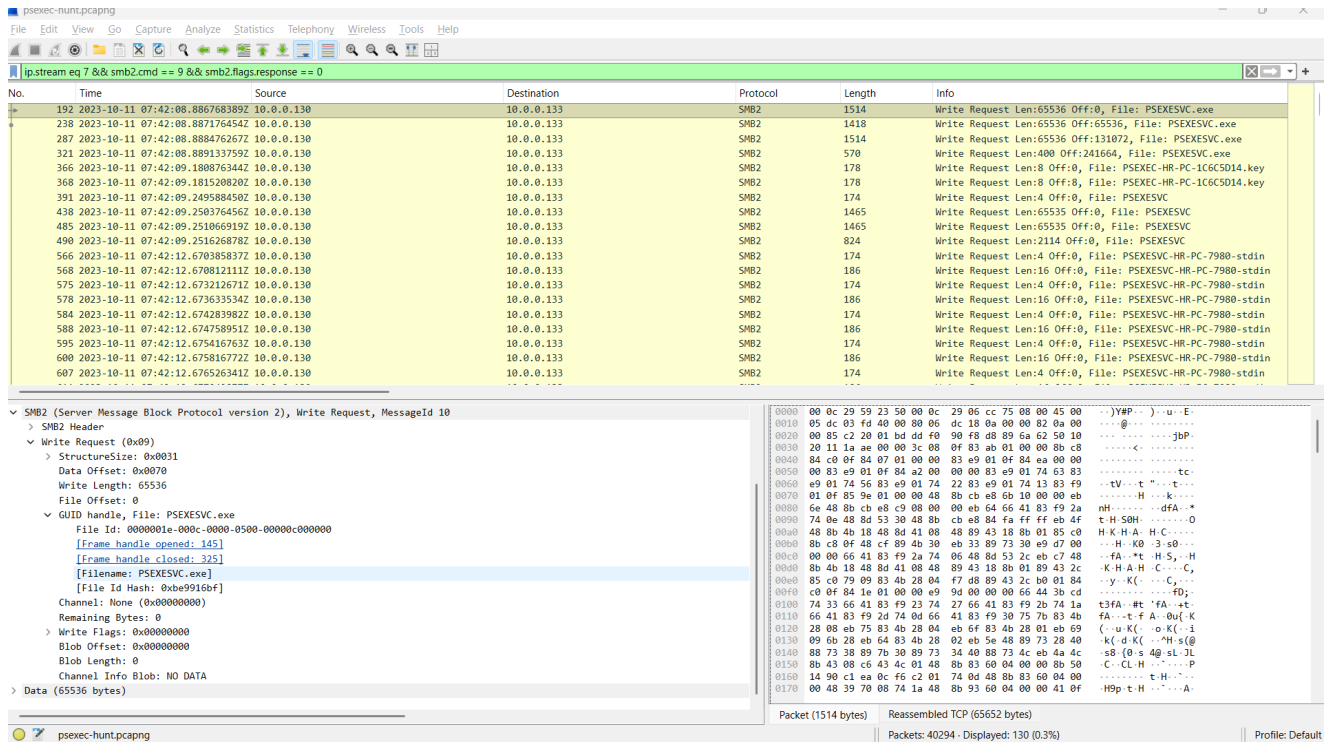


Figure 9. SMB2 Write requests transferring PSEXESVC.exe.

PsExec is a legitimate Microsoft Sysinternals remote-administration utility. Its use becomes incident-relevant through the surrounding unauthorized behavior chain.

4.7 Remote Service Control

After the file transfer, DCE/RPC traffic exposes SVCCTL calls to the remote Windows Service Control Manager.

```
ip.stream eq 7 && dcerpc
```

Operation	Analytical meaning
OpenSCManager2	Open a handle to the Service Control Manager.
CreateServiceW	Register a service on the target.
OpenServiceW	Open the newly created or existing service.
StartServiceW	Start the service and execute its binary.
QueryServiceStatus	Check the resulting service state.

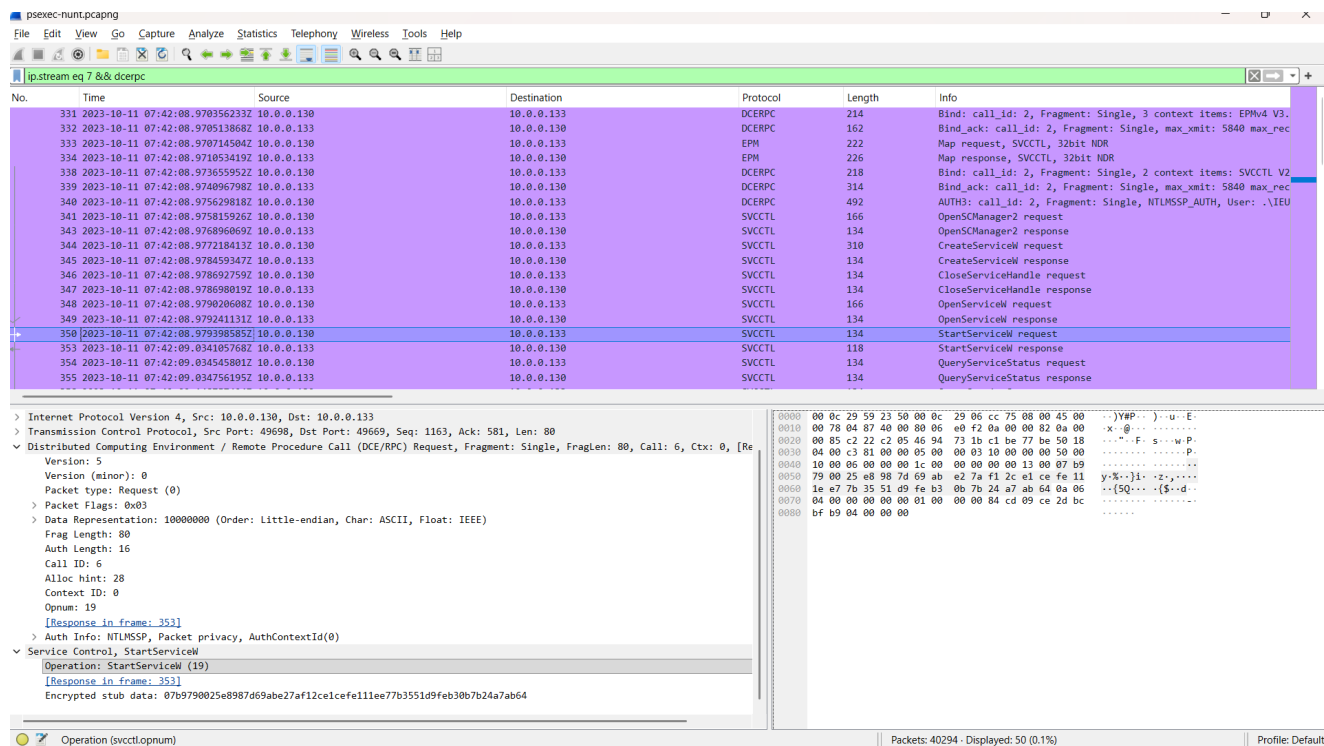


Figure 10. SVCCTL StartServiceW and related remote service-control operations.

The sequence ties the transferred executable to remote execution; PSEXESVC.exe was not merely stored on SALES-PC.

4.8 PsExec Named Pipes

After service start, PsExec opens named pipes for standard input, standard output, and standard error.

ip.stream eq 7 && smb2.filename contains "PSEXESVC-"

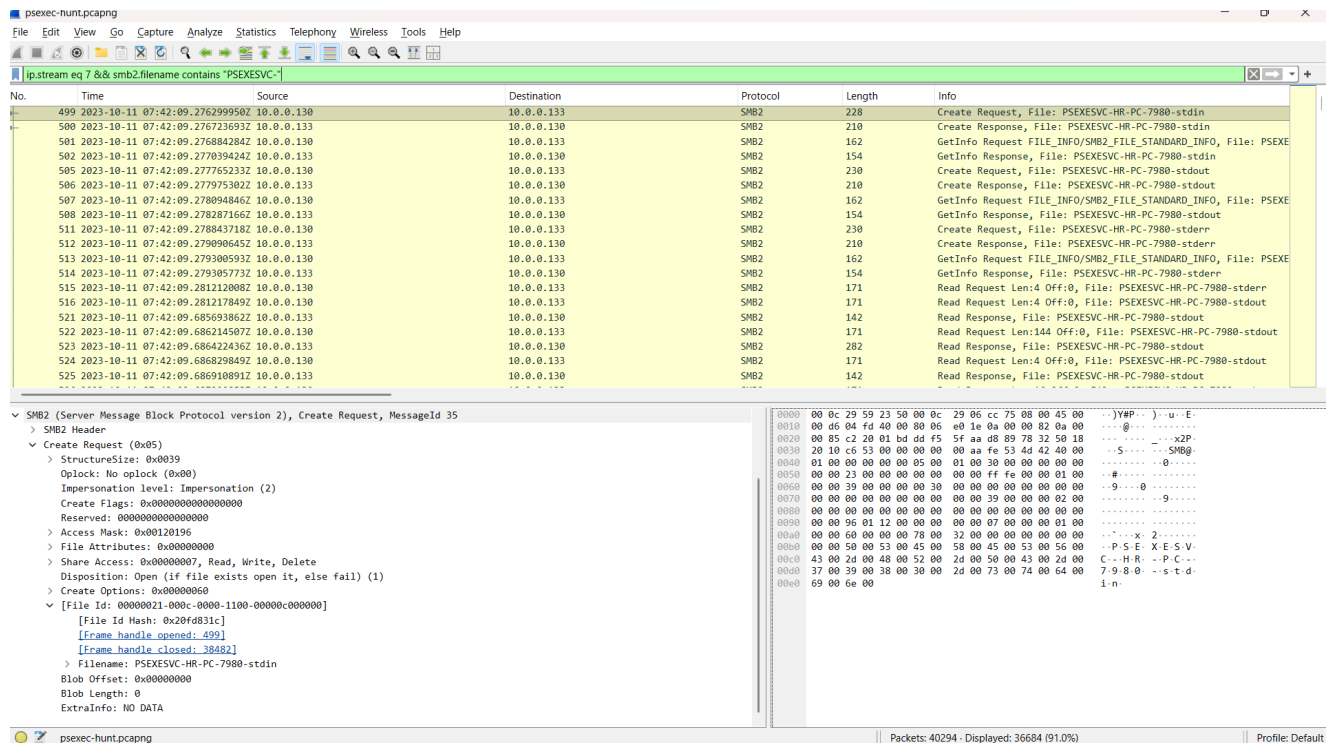


Figure 11. PsExec stdin, stdout, and stderr named pipes.

Named pipe	Function
PSEXESVC-HR-PC-7980-stdin	Carries input or command data to the remote process.
PSEXESVC-HR-PC-7980-stdout	Returns normal process output.
PSEXESVC-HR-PC-7980-stderr	Returns error output.

HR-PC identifies the initiating workstation. The value 7980 is the PsExec client process identifier used to distinguish this set of pipes.

4.8.1 Named-Pipe Payload Limitation

Read responses for the stdout pipe contain data, but the selected 144-byte payload is not readable as plain command output in this capture.

```
ip.stream eq 7 && smb2.cmd == 8 &&
offset_length(smb2.file_data, 0, 144) &&
smb2.filename contains "stdout"
```

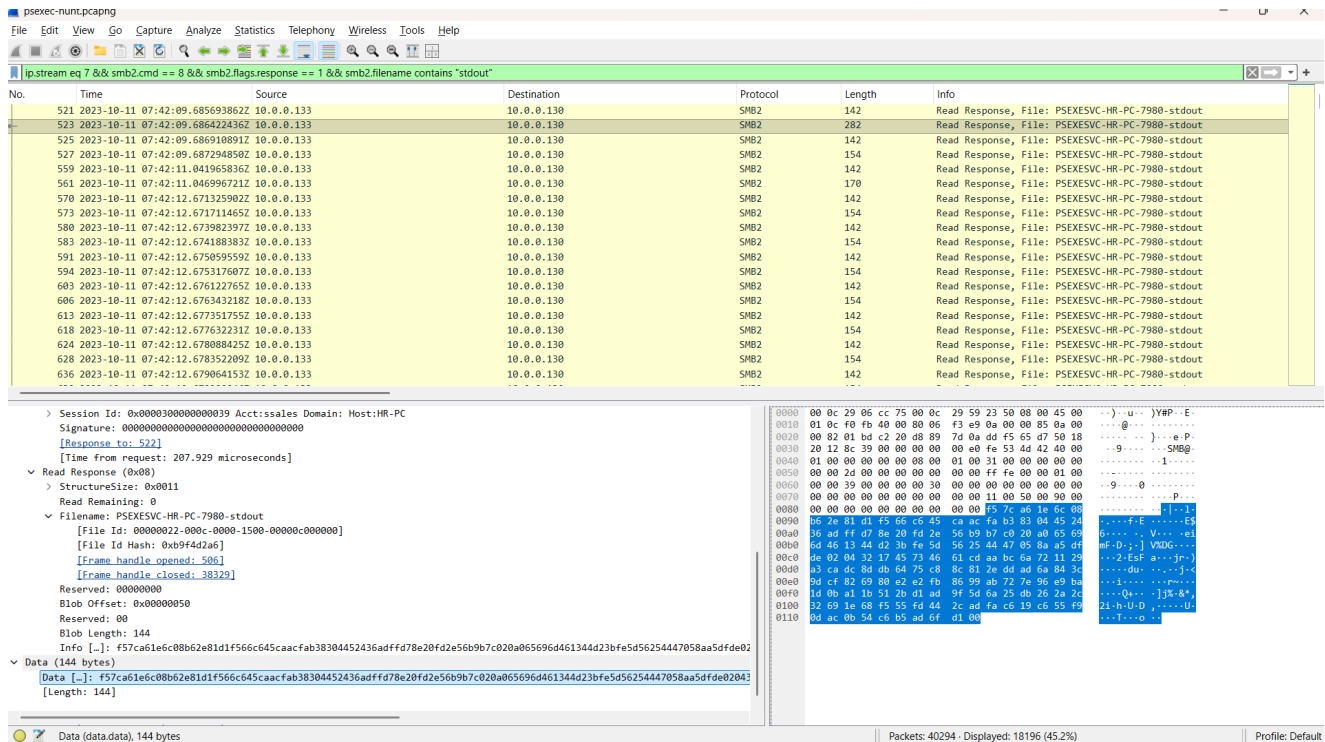


Figure 12. Opaque data returned through the PsExec stdout named pipe.

Defensible conclusion: command output was exchanged through the pipe. The exact command and output cannot be recovered reliably from this PCAP alone.

4.9 Second Pivot Target

Initial TCP SYN packets to destination port 445 reveal later SMB connection attempts from the same source.

```
ip.src == 10.0.0.130 &&
tcp.dstport == 445 &&
tcp.flags.syn == 1 &&
tcp.flags.ack == 0
```

The image shows a Wireshark capture of network traffic. The top pane displays a list of packets with a filter applied: `ip.src == 10.0.0.130 && tcp.dstport == 445 && tcp.flags.syn == 1 && tcp.flags.ack == 0`. The middle pane shows the packet details for the selected packet (No. 38506), including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The bottom pane shows the packet bytes in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
123	2023-10-11 07:42:08.8468282292	10.0.0.130	10.0.0.133	TCP	66	49696 → 445 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
38506	2023-10-11 07:46:19.8824735712	10.0.0.130	10.0.0.131	TCP	66	49701 → 445 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
38528	2023-10-11 07:46:21.9725945672	10.0.0.130	10.0.0.131	TCP	66	49703 → 445 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
39105	2023-10-11 07:46:33.5590894782	10.0.0.130	10.0.0.131	TCP	66	49707 → 445 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
39693	2023-10-11 07:46:40.0313239632	10.0.0.130	10.0.0.131	TCP	66	49711 → 445 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM

Packet details for No. 38506:

- Frame 38506: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface ens33, id 0
- Ethernet II, Src: VMware_06:cc:75 (00:0c:29:06:cc:75), Dst: VMware_84:0f:8f (00:0c:29:84:0f:8f)
- Internet Protocol Version 4, Src: 10.0.0.130, Dst: 10.0.0.131
- Transmission Control Protocol, Src Port: 49701, Dst Port: 445, Seq: 0, Len: 0

Packet bytes (hex):

```
0000 00 0c 29 84 0f 8f 00 0c 29 06 cc 75 08 00 45 00 ... ..E..
0010 00 34 19 22 40 00 80 06 cc 9d 0a 00 00 82 0a 00 ..4..@...
0020 00 83 c2 25 01 bd 05 dc 91 0f 00 00 00 00 02 ...S...
0030 fa f0 04 4d 00 00 02 04 05 b4 01 03 03 08 01 01 ...M...
0040 04 02 ..
```

Figure 13. Initial SMB SYN packets to 10.0.0.133 and the later target 10.0.0.131.

The first target appears at approximately 07:42:08 UTC. Beginning around 07:46:19 UTC, 10.0.0.130 initiated multiple TCP/445 connections to 10.0.0.131.

The second server's NTLMSSP_CHALLENGE identifies its hostname through Target Info.

```
ip.src == 10.0.0.131 && ntlmssp.message.type == 2
```

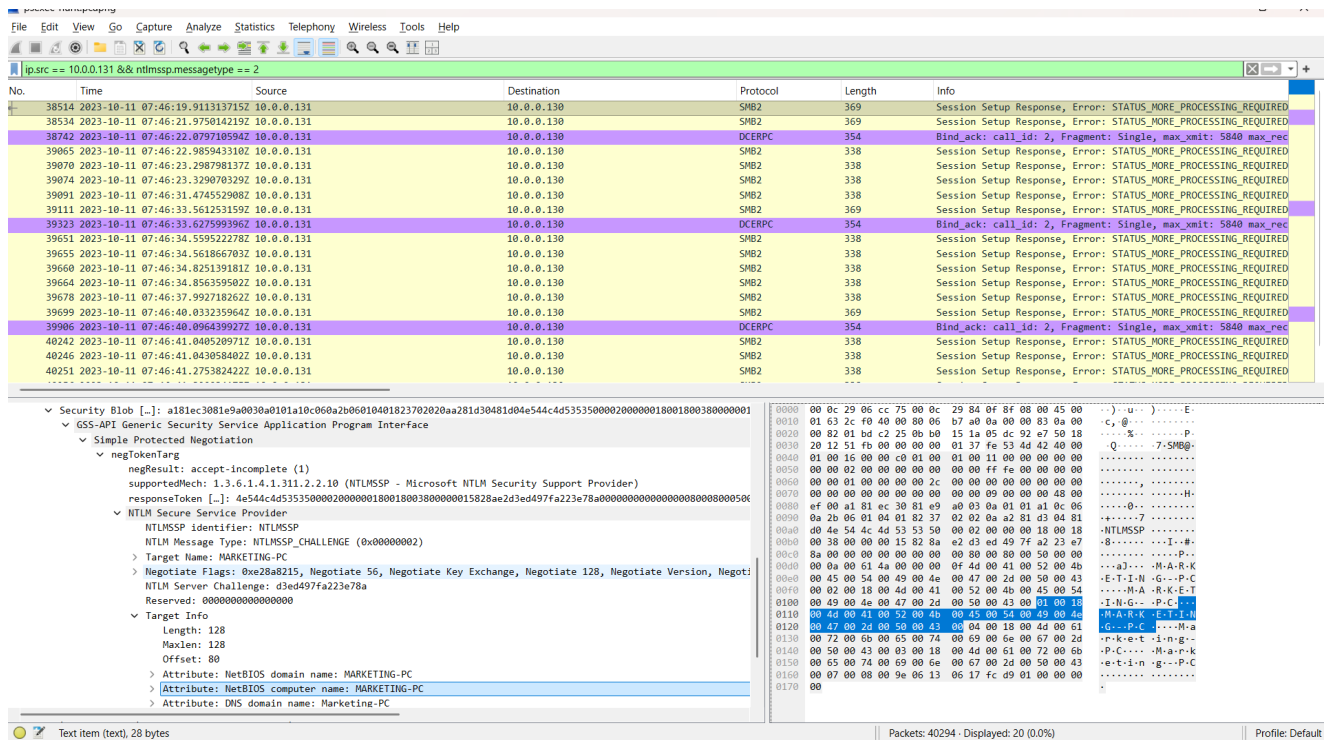


Figure 14. NTLM challenge identifying 10.0.0.131 as MARKETING-PC.

5. Reconstructed Timeline

Time (UTC)	Source	Destination	Observed activity
2023-10-11 07:42:08	10.0.0.130	10.0.0.133	SMB connection and negotiation
2023-10-11 07:42:08	HR-PC	SALES-PC	NTLM authentication using ssaes
2023-10-11 07:42:08	HR-PC	SALES-PC	IPC\$ and ADMIN\$ access
2023-10-11 07:42:08	HR-PC	SALES-PC	PSEXESVC.exe transfer
2023-10-11 07:42:08	HR-PC	SALES-PC	Service creation and start
2023-10-11 07:42:09	HR-PC	SALES-PC	PsExec named-pipe exchange
2023-10-11 07:46:19	10.0.0.130	10.0.0.131	New SMB sessions to MARKETING-PC

6. Answer and Evidence Matrix

Question	Answer	Primary evidence
Q1 — Initial source IP	10.0.0.130	Client SMB2 NEGOTIATE requests
Q2 — First pivot hostname	SALES-PC	NTLM challenge Target Info
Q3 — Authentication username	ssales	NTLMSSP_AUTH user field
Q4 — Service executable	PSEXESVC.exe	SMB2 Create and Write
Q5 — Installation share	ADMIN\$	SMB2 Tree Connect
Q6 — Communication share	IPC\$	Tree Connect, RPC, and named pipes
Q7 — Second pivot hostname	MARKETING-PC	Second target NTLM challenge

7. Indicators of Compromise

Type	Value
Source endpoint	10.0.0.130 / HR-PC
First target	10.0.0.133 / SALES-PC
Second target	10.0.0.131 / MARKETING-PC
User account	ssales
Executable	PSEXESVC.exe
Shares	ADMIN\$, IPC\$
Named-pipe prefix	PSEXESVC-HR-PC-7980-
Service	SMB over TCP/445

8. MITRE ATT&CK Mapping

Tactic	Technique	Evidence
Lateral Movement	T1021.002 — SMB/Windows Admin Shares	SMB authentication and ADMIN\$/IPC\$ access
Lateral Movement	T1570 — Lateral Tool Transfer	PSEXESVC.exe copied to SALES-PC
Execution	T1569.002 — Service Execution	CreateServiceW and StartServiceW
Multiple	T1078 — Valid Accounts	Successful NTLM use of ssales

9. Detection Opportunities

- Correlate successful NTLM logons with immediate access to ADMIN\$ or IPC\$.
- Alert on writes of PSEXESVC.exe or similarly named service binaries to administrative shares.
- Detect remote service creation shortly after an SMB file transfer.
- Monitor SVCCTL operations such as CreateServiceW and StartServiceW from workstations.
- Alert on named pipes matching PSEXESVC*-(stdin|stdout|stderr).
- Correlate Security events 4624, 4672, 4697, 5140, and 5145 with System event 7045.
- Flag one workstation initiating SMB sessions to multiple peer workstations within a short interval.
- Review endpoint process trees for services.exe starting unexpected binaries or shells.

10. Remediation Recommendations

7. Isolate HR-PC, SALES-PC, and MARKETING-PC pending endpoint validation.
8. Reset ssales credentials and review the account's privileges, group memberships, and logon history.
9. Restrict SMB between workstation segments with host and network firewalls.
10. Limit ADMIN\$ and IPC\$ access to approved administrative systems and accounts.
11. Reduce or disable NTLM where operationally possible and enforce modern authentication controls.
12. Control PsExec and other dual-use remote-administration utilities through application control and monitoring.
13. Hunt for the PSEXESVC service, executable remnants, related named pipes, and service-creation events.
14. Collect EDR, Active Directory, Windows event, and process telemetry to determine the original compromise path and executed commands.

11. Limitations

- The investigation is limited to the supplied PCAP; no endpoint, Active Directory, or EDR data was available.
- The NTLM exchange confirms that ssales was used but does not establish how the credentials were acquired.
- The visible PsExec pipe payload is opaque, so exact remote commands and outputs are not asserted.
- PSEXESVC.exe is a dual-use artifact. Classification depends on the correlated unauthorized behavior sequence.

12. Conclusion

The packet evidence confirms PsExec-based lateral movement from HR-PC (10.0.0.130) to SALES-PC (10.0.0.133) using the ssales account. The source connected to IPC\$ and ADMIN\$, transferred PSEXESVC.exe, controlled a remote service through SVCCTL, and exchanged data through PsExec named pipes. The same source subsequently opened SMB sessions to MARKETING-PC (10.0.0.131), showing that the activity extended beyond the first target.

The strongest analytical result is the correlation of multiple protocol artifacts rather than reliance on a single filename or packet. Authentication identity, share access, executable transfer, service execution, and named-pipe communication collectively reconstruct the lateral-movement workflow.

13. References

15. Microsoft Sysinternals — PsExec: <https://learn.microsoft.com/en-us/sysinternals/downloads/psexec>
16. MITRE ATT&CK — PsExec (S0029): <https://attack.mitre.org/software/S0029/>
17. MITRE ATT&CK — SMB/Windows Admin Shares (T1021.002): <https://attack.mitre.org/techniques/T1021/002/>
18. MITRE ATT&CK — Service Execution (T1569.002): <https://attack.mitre.org/techniques/T1569/002/>
19. MITRE ATT&CK — Lateral Tool Transfer (T1570): <https://attack.mitre.org/techniques/T1570/>
20. MITRE ATT&CK — Valid Accounts (T1078): <https://attack.mitre.org/techniques/T1078/>

Final assessment: The environment should be treated as compromised until the involved endpoints, account activity, and service-execution telemetry are validated and remediated.